

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Subject

Cybersecurity Law & Policy

Project Type

Group / Individual Project

Total Marks

100 Marks

DeliverablesReport + Presentation + Flowcharts +
Infographics**Important Notice to Students**

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

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PART 1: Introduction

Prepare an overview section that sets the foundation for your cybersecurity law. This section should clearly articulate why cybersecurity legislation is necessary for VIT's.

What to Include

- Purpose of the law — What is this law designed to achieve?
- Importance of cybersecurity — Why does VIT's need this law now?
- Challenges faced by VIT's — Describe specific threats from the scenario
- Goals and expected outcomes — What will change after this law is enacted?

Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyberbullying incidents.

PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

#	Term	Definition (To be Completed by the Student)
1	Cybercrime	Any criminal activity that either targets a computer, a computer network, or a networked device, or uses them as the primary instrument to commit a crime is Known as Cybercrime.
2	Hacker	A person who uses technical skills to get unauthorized access to computer systems, networks, or data. In a legal context, it typically refers to malicious actors (black-hat hackers) breaking into systems without authorization is a Hacker.
3	Ethical Hacker	An authorized cybersecurity expert who systematically attempts to penetrate a computer system or network with the owner's permission to find and fix security vulnerabilities is an Ethical Hacker.
4	Malware	Malware is an malicious software which is designed to cause damage, disruption and unauthorized access to a computer, server, client, or computer network .
5	Ransomware	It is a type of Malware that encrypts victim's files and ask for payment in bitcoins for decryption key for accessing the files
6	Phishing	It is a type of social engineering attack in which attackers send fraud emails to many people which is designed to look like they come from a reputable source, aiming to trick individuals into revealing sensitive information like passwords or credit card numbers.
7	Data Breach	A security threat in which sensitive, protected and confidential data is stolen or used by an individual unauthorized to do so which triggers legal reporting liabilities under privacy regulations.
8	Personal Data	Any information relating to an identified person this includes direct identifiers like names and ID numbers, as well as indirect identifiers like location data or online identifiers.
9	Sensitive Data	A specific part of personal data which requires a higher level of legal protection due to the severe risks of discrimination or harm if exposed.
10	Digital Signature	Digital Signature is used to validate the authenticity and integrity of a digital message, document. It links a signer to a document in a legally binding manner and ensures non-repudiation.
11	Electronic Evidence	Any probative information stored or transmitted in digital form that a party in a court case can use at trial to support or refute a legal claim is known as Electronic Evidence or Digital Evidence.
12	Identity Theft	The fraudulent use of other individual's personal identifying information usually to commit other crimes under a false identity

13	Artificial Intelligence	The simulation of human intelligence processes by machines, especially computer systems is Artificial Intelligence . In cybersecurity, AI is used to detect anomalies and threats in real-time basis.
14	Cloud Computing	The on-demand delivery of computing services which includes data storage, servers, databases, networking and software over the internet shifts data management responsibilities to a shared responsibility model between the user and the cloud provider.
15	Critical Infrastructure	The physical and virtual assets, systems, and networks which are vital to a nation that their incapacitation or destruction would have a debilitating impact on national security, economic security, public health or safety.
16	Cyberbullying	The use of digital communication to harass, threaten, humiliate or target another person often carrying distinct civil or criminal penalties depending on jurisdiction.
17	Deepfake	Synthetic media in which a person in an existing image, audio recording or video is replaced or altered with someone else's likeness using advanced machine learning and artificial intelligence techniques which are often used for misinformation and identity fraud.
18	Encryption	The process of converting legible information into an unreadable, scrambled format using a cryptographic algorithm and key. Only authorized parties with the correct decryption key can reverse the process to read the data.
19	Firewall	A network security software that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies acting as a barrier between a trusted network and untrusted networks.
20	Zero-Day Exploit	A cyberattack that targets a software or hardware vulnerability which is completely unknown to the vendor. Because the vulnerability is discovered by attackers before a patch or fix can be created, there are "zero days" of protection available against it .

PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy

Citizens must be protected from unauthorized collection, use, or sharing of their personal data. Every citizen has the right to privacy and protection of personal information. No organization or government agency may collect, use, or share personal data without lawful purpose and informed consent.

Responsible Authority: Data Protection Officers (DPOs) and NCSA.

Enforcement: Violations may result in fines up to ₹5 crore and corrective action by the NCSA.

Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures.

Citizens have the right to expect that their personal data is protected using appropriate security measures such as encryption, access controls, and regular security audits.

Responsible Authority: Organizations handling personal data and NCSA.

Enforcement: Negligence leading to data breaches may result in penalties, mandatory audits, and compensation to affected individuals.

Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems.

Citizens may request deletion of their personal information when it is no longer required or when consent has been withdrawn, unless retention is required by law.

Responsible Authority: Data Controllers and Data Protection Officers.

Enforcement: Requests must be processed within 30 days. Failure to comply may result in fines.

Right to Information

Citizens have the right to know what data is collected about them and how it is used.

Citizens have the right to know what personal data is collected, how it is used, and with whom it is shared. Organizations must provide clear privacy notices.

Responsible Authority: Organizations collecting personal data.

Enforcement: Information requests must be answered within 15 working days or penalties may apply.

Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes.

Citizens have the right to report cybercrimes such as phishing, identity theft, cyberbullying, online fraud, and ransomware attacks without fear of retaliation.

Responsible Authority: Cyber Crime Police Units and NCSA.

Enforcement: Authorities must acknowledge complaints within 24 hours and begin investigation promptly.

Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation.

Every citizen affected by or accused of a cybercrime is entitled to a fair, transparent, and timely investigation based on legally obtained digital evidence.

Responsible Authority: Cyber Crime Investigation Units, Digital Forensics Labs, and Courts.

Enforcement: Investigations should be completed within a reasonable time and citizens may appeal unfair decisions.

Right to Digital Education and Awareness

Citizens have the right to receive cybersecurity education and awareness training to help them stay safe online.

Responsible Authority: Ministry of Education, NCSA, schools, universities, and employers.

Enforcement: Educational institutions and large organizations must conduct regular cybersecurity awareness programs.

Instruction

You must include at least 6 rights. You may add more. Each right should be at least 2-3 paragraphs with specific implementation details.

[Student: Expand this right with specific provisions, who is responsible, and how it will be enforced.]



PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + VIT's Example	Punishment	Fine
Unauthorized Access	Accessing a government portal without permission	Up to 3 years imprisonment	₹5 lakh
Data Theft	Stealing citizen or company data	Up to 5 years imprisonment	₹10 lakh
Identity Theft	Using another person's digital identity	Up to 5 years imprisonment	₹10 lakh
Phishing	Fake bank/email scams targeting citizens	Up to 4 years imprisonment	₹8 lakh
Ransomware	Encrypting systems and demanding payment	Up to 10 years imprisonment	₹25 lakh
Malware Distribution	Creating or spreading malicious software	Up to 7 years imprisonment	₹15 lakh
Cyberbullying	Online harassment causing harm	Up to 2 years imprisonment	₹2 lakh
Online Harassment	Threats, stalking, or abusive communication	Up to 3 years imprisonment	₹5 lakh
Financial Fraud	Online banking or payment fraud	Up to 7 years imprisonment	₹20 lakh
Misinformation	Deliberate spread of false information causing public harm	Up to 3 years imprisonment	₹5 lakh
Website Defacement	Unauthorized alteration of websites	Up to 5 years imprisonment	₹10 lakh
Cyber Terrorism	Attacks on critical infrastructure or national security	Up to Life Imprisonment	₹1 crore

PART 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. Organizations shall collect only the minimum personal data necessary for a specific and lawful purpose.
2. Personal data shall not be collected through deceptive, misleading, or unauthorized methods.
3. Explicit user consent must be obtained before collecting sensitive personal data, including biometric, health, financial, and location information.
4. Organizations must provide a privacy notice explaining the purpose of collection, storage period, and data-sharing practices.
5. Children's data (under 18 years) shall require parental or guardian consent before collection.

Data Storage Rules

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. All personal and sensitive data shall be stored using industry-standard encryption methods.
2. Organizations shall implement Multi-Factor Authentication (MFA) for systems containing sensitive information.
3. Access to personal data shall be restricted based on job responsibilities and the principle of least privilege.
4. Regular security audits and vulnerability assessments shall be conducted at least once every year.
5. Backup copies of critical data shall be maintained securely and protected against unauthorized access.

Data Sharing Rules

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. Personal data shall not be shared with third parties without the individual's consent unless required by law.
2. Third-party organizations receiving personal data must maintain equivalent security standards.
3. Organizations shall maintain records of all data-sharing activities.
4. Cross-border transfer of personal data shall only be permitted to countries with adequate data protection standards.
5. Shared data shall be limited to the minimum amount required for the intended purpose.

User Consent Requirements

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. Consent must be freely given, informed, specific, and unambiguous.
2. Consent requests shall be presented in clear and understandable language.
3. Users shall have the right to withdraw consent at any time without penalty.
4. Organizations must maintain records proving that valid consent was obtained.
5. Separate consent shall be required for marketing, profiling, and third-party data sharing.

Data Retention Period

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. Personal data shall be retained only for the period necessary to fulfill the purpose for which it was collected.
2. Financial and legal records may be retained for up to 7 years where required by law.
3. Expired or unnecessary data must be securely deleted or anonymized.
4. Organizations shall maintain documented retention schedules for all categories of data.
5. Citizens shall be informed of applicable retention periods.

Data Deletion Process

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. Citizens may request deletion of their personal data when it is no longer required or when consent has been withdrawn.
2. Organizations must process valid deletion requests within 30 days.
3. Deleted data must be permanently removed from active systems and databases.
4. Data retained for legal or regulatory obligations shall be exempt until the retention period expires.
5. Organizations shall provide confirmation once deletion has been completed.

Breach Notification Requirements

[Student: Write detailed rules for this section. Include specific technical and legal requirements.]

1. Any data breach affecting personal information must be reported to the National Cyber Security Authority (NCSA) within 72 hours of discovery.
2. Affected individuals shall be notified without undue delay if the breach presents a significant risk.
3. Organizations must disclose the nature of the breach, affected data categories, potential impact, and corrective actions taken.
4. Incident response teams shall investigate the cause of the breach and implement preventive measures.
5. Failure to report a breach within the prescribed period may result in fines ranging from ₹10 lakh to ₹5 crore depending on the severity of the incident.

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: Data Protection

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

All organizations shall protect customer and employee data using appropriate security measures, including encryption and access controls.

Applies to: All organizations handling personal data.

Penalty: Fine up to ₹50 lakh for negligence resulting in data breaches.

Responsibility 2 : Security Audits

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall conduct cybersecurity audits at least once every year to identify and address vulnerabilities.

Applies to: Medium and large organizations.

Penalty: Fine up to ₹25 lakh for failure to conduct audits.

Responsibility 3: Employee Cybersecurity Training

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall provide cybersecurity awareness training to employees at least annually.

Applies to: All organizations.

Penalty: Fine up to ₹10 lakh for non-compliance

Responsibility 4: Incident Reporting

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall report significant cybersecurity incidents to the National Cyber Security Authority (NCSA) within 72 hours.

Applies to: All organizations.

Penalty: Fine up to ₹1 crore for failure to report.

Responsibility 5: Appointment of Data Protection Officer (DPO)

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations processing large amounts of personal data shall appoint a qualified Data Protection Officer.

Applies to: Government agencies and large enterprises.

Penalty: Fine up to ₹20 lakh.

Responsibility 6: Access Control Management

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall implement role-based access controls and Multi-Factor Authentication (MFA) for critical systems.

Applies to: All organizations.

Penalty: Fine up to ₹25 lakh.

Responsibility 7: Data Backup and Recovery

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall maintain secure backups and disaster recovery plans to ensure business continuity.

Applies to: Critical infrastructure and large organizations.

Penalty: Fine up to ₹30 lakhs

Responsibility 8: Vulnerability Assessment and Penetration Testing

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall conduct annual vulnerability assessments and penetration testing of critical systems.

Applies to: Financial institutions, healthcare providers, and critical infrastructure operators.

Penalty: Fine up to ₹25 lakh.

Responsibility 9: Secure Software Development

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations developing software shall follow secure coding practices and promptly address security vulnerabilities.

Applies to: Software companies and technology providers.

Penalty: Fine up to ₹20 lakh.

Responsibility 10 : Incident Response Plan

[Student: Describe the specific obligation, who it applies to, and the consequence for non-compliance.]

Organizations shall maintain a documented Cyber Incident Response Plan and conduct regular response drills.

Applies to: All organizations.

Penalty: Fine up to ₹25 lakh.

Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity | Report incidents within 72 hours | Protect customer information | Appoint a Data Protection Officer | Conduct annual penetration testing | Implement access control policies | Encrypt sensitive data | Maintain incident response plans

PART 7: Government Responsibilities

Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

Protect National Infrastructure

[Hint: Define which infrastructure is critical and how it must be protected.]

The government shall identify and protect critical infrastructure, including banks, hospitals, power plants, transportation systems, telecommunications networks, and government data centers. Critical infrastructure operators must implement encryption, continuous monitoring, annual security audits, and disaster recovery plans.

The NCSA shall conduct periodic inspections and issue security directives when necessary.

Monitor Cyber Threats

[Hint: Describe surveillance systems, threat intelligence gathering, and monitoring protocols.]

The government shall establish a National Cyber Threat Monitoring Center to monitor cyber threats, collect threat intelligence, and coordinate responses to cyber incidents. Advanced monitoring systems, Security Operations Centers (SOCs), and AI-based threat detection tools shall be used to identify and mitigate cyber threats in real time.

Investigate Cybercrimes

[Hint: Outline the legal processes for cyber investigations including warrants and digital forensics.]

The government shall establish specialized Cyber Crime Investigation Units and Digital Forensics Laboratories. Cybercrime investigations shall follow legal procedures, preserve digital evidence, and obtain judicial warrants where required. Serious cyber offenses shall be investigated jointly by law enforcement and the NCSA.

International Coordination

[Hint: Explain how VIT's will work with other countries' cyber agencies and INTERPOL.]

The government shall cooperate with foreign governments, CERT organizations and INTERPOL to combat cross-border cybercrime. Information sharing agreements, joint investigations, extradition procedures, and cyber threat intelligence exchanges shall be established to strengthen international cybersecurity cooperation.

Awareness Programs

[Hint: Describe government-led public education campaigns about cybersecurity.]

The government shall conduct nationwide cybersecurity awareness campaigns through schools, universities, workplaces, television, radio, and social media. Programs shall educate citizens on phishing, ransomware, online fraud, password security, digital privacy, and responsible internet usage. Annual Cybersecurity Awareness Week shall be organized to promote cyber hygiene and digital safety.

PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

Reporting Process Steps

Step	Stage	Description
1	Incident Detected	Employees, users, or monitoring systems identify a cybersecurity incident such as phishing, malware, ransomware, unauthorized access, or data breach. The incident must be reported internally within 24 hours.
2	Internal Investigation	The organization's Incident Response Team collects logs, preserves evidence, identifies affected systems, and determines the nature and scope of the incident.
3	Report to Authority	Significant incidents must be reported to the National Cyber Security Authority (NCSA) within 72 hours. Critical infrastructure incidents must be reported immediately.
4	Risk Assessment	NCSA and the organization assess the severity, affected users, business impact, data exposure, and potential national security risks.
5	Mitigation	Immediate actions are taken to contain the threat, isolate affected systems, block malicious activity, patch vulnerabilities, and prevent further damage.
6	Recovery	Systems and services are restored using secure backups, security controls are strengthened, and operations return to normal while continuous monitoring is maintained.
7	Final Report	A detailed report is submitted to NCSA including root cause analysis, impact assessment, corrective actions, lessons learned, and recommendations to prevent recurrence.

Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.

PART 9: National Cyber Security Authority (NCSA)

Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

Organizational Structure

[Hint: Define departments, leadership roles (Director General, Deputy Directors, etc.), and reporting lines.]

Director General (DG) – Head of NCSA

- Deputy Director General
- Threat Intelligence Division
- Incident Response Division
- Digital Forensics Division
- Legal & Compliance Division
- Public Awareness & Training Division

Regional Cybersecurity Offices shall operate in all major provinces and report to the Deputy Director General.

Core Responsibilities

[Hint: List all primary functions — from policy-making to public education.]

1. Develop national cybersecurity policies and standards.
2. Protect critical national infrastructure.
3. Monitor and respond to cyber threats and incidents.
4. Coordinate national cyber defense activities.
5. Conduct cybersecurity audits and compliance inspections.
6. Investigate major cybercrime incidents.
7. Promote cybersecurity awareness and education.
8. Facilitate international cybersecurity cooperation.
9. Issue cybersecurity advisories and alerts.
10. Support digital forensics and cybercrime investigations.

Legal Powers

[Hint: What powers does the NCSA have? Can it compel companies to share data? Can it shut down websites?]

1. Require organizations to report significant cyber incidents.
2. Conduct cybersecurity inspections and audits.
3. Issue mandatory security directives.
4. Order temporary shutdown of compromised systems posing national security risks.
5. Request information relevant to cybercrime investigations.
6. Impose administrative fines for non-compliance.
7. Coordinate with law enforcement agencies for criminal investigations.
8. Recommend blocking malicious websites, domains, infrastructure through judicial authorization.

Investigation Process

[Hint: Step-by-step: How does the NCSA investigate a cybercrime complaint?]

1. Receive cybercrime complaint or incident report.
2. Conduct preliminary assessment and classify severity.
3. Collect digital evidence and system logs.
4. Perform forensic investigation and threat analysis.
5. Coordinate with law enforcement if criminal activity is detected.
6. Identify responsible individuals or organizations.
7. Issue enforcement actions, penalties, or prosecution recommendations.
8. Publish lessons learned and preventive guidance.

Departments

[Hint: Define at least 5 specialized departments (e.g., Threat Intelligence Unit, Incident Response Team, Digital Forensics Lab).]

1. Threat Intelligence Unit
Monitors cyber threats, gathers intelligence, and issues national security alerts.
2. National Incident Response Team (NIRT)
Handles major cyber incidents, ransomware attacks, and national cyber emergencies.
3. Digital Forensics Laboratory
Collects, preserves, and analyzes digital evidence for investigations.
4. Legal & Compliance Division
Ensures compliance with cybersecurity laws and conducts regulatory enforcement.
5. Critical Infrastructure Protection Division
Protects banks, hospitals, power plants, telecommunications, and government systems.
6. Cyber Awareness & Training Division
Conducts public awareness campaigns and cybersecurity education programs.

PART 10: AI and Emerging Technology Regulations

Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

Artificial Intelligence

Key Risks

[List specific cybersecurity risks this technology poses in VIT's]

- AI-generated cyberattacks and malware.
- Algorithmic bias and discrimination.
- Privacy violations through excessive data collection.
- Deepfake generation and misinformation.

Regulations

[Write specific rules that govern the use of this technology]

- High-risk AI systems must undergo security and compliance audits.
- AI decisions affecting citizens must be explainable and transparent.
- Organizations shall conduct annual AI risk assessments.
- AI systems processing personal data must comply with privacy laws.

Deepfakes

Key Risks

[List specific cybersecurity risks this technology poses in VIT's]

- Identity theft and impersonation.
- Election interference and misinformation.
- Financial fraud and social engineering attacks.
- Reputational damage.

Regulations

[Write specific rules that govern the use of this technology]

- AI-generated content must contain visible disclosure or watermarking.
- Creating malicious deepfakes without consent is prohibited.
- Deepfakes used for fraud or harassment shall attract criminal penalties.
- Platforms must remove unlawful deepfake content upon notification.

IoT Devices

Key Risks

[List specific cybersecurity risks this technology poses in VIT's]

- Weak passwords and insecure configurations.
- Unauthorized device access.
- Large-scale botnet attacks.
- Data leakage from connected devices.

Regulations

[Write specific rules that govern the use of this technology]

- Default passwords must be changed before deployment.
- Manufacturers shall provide timely security updates.
- Sensitive IoT communications must be encrypted.
- Annual security testing is mandatory for critical IoT devices.

Cloud Computing

Key Risks

[List specific cybersecurity risks this technology poses in VIT's]

- Data breaches and unauthorized access.
- Misconfigured cloud environments.
- Data loss and service disruption.
- Cross-border data transfer risks.

Regulations

[Write specific rules that govern the use of this technology]

- Cloud providers must encrypt data at rest and in transit.
- Multi-Factor Authentication (MFA) is mandatory.
- Organizations shall maintain backup and disaster recovery plans.
- Cross-border data transfers require adequate legal protection.

Digital Identity Systems

Key Risks

[List specific cybersecurity risks this technology poses in VIT's]

- Identity theft and credential compromise.
- Biometric data misuse.
- Unauthorized account access.
- Centralized database breaches.

Regulations

[Write specific rules that govern the use of this technology]

- MFA shall be mandatory for digital identity systems.
- Biometric data must be encrypted and securely stored.
- Citizens shall have access to identity activity logs.
- Regular security audits shall be conducted.

Autonomous Systems

Key Risks

[List specific cybersecurity risks this technology poses in VIT's]

- System malfunction causing physical or financial harm.
- Unauthorized control by cyber attackers.
- AI decision-making errors.
- Safety failures in critical operations.

Regulations

[Write specific rules that govern the use of this technology]

- Autonomous systems must undergo security and safety certification before deployment.
- Human oversight shall be maintained for high-risk operations.
- Emergency shutdown mechanisms must be available.
- Operators shall be legally responsible for negligence resulting from system failures.

PART 11: Critical Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

Sector	Cybersecurity requirements
Banks & Financial	1. Multi-Factor Authentication (MFA) for all transactions. 2. End-to-end encryption of customer data. 3. Real-time fraud detection systems. 4. Annual penetration testing. 5. Mandatory incident reporting within 24 hours.
Hospitals & Healthcare	1. Encryption of electronic health records. 2. Restricted access to patient data. 3. Regular backup of medical systems. 4. Continuous monitoring of medical devices. 5. Emergency cyber incident response plan.
Airports & Aviation	1. Secure access controls for operational systems. 2. Network segmentation between public and critical systems. 3. Continuous threat monitoring. 4. Regular vulnerability assessments. 5. Disaster recovery and business continuity plans.
Railways & Transport	1. Protection of signaling and control systems. 2. Real-time monitoring of transport networks. 3. Secure communication channels. 4. Regular cybersecurity audits. 5. Incident response and recovery procedures.
Power Plants & Energy	1. Isolation of operational technology (OT) networks. 2. 24/7 Security Operations Center (SOC) monitoring. 3. Strict access control policies. 4. Regular patch management. 5. Backup control systems and disaster recovery plans.
Government Data Centers	1. Encryption of all sensitive government data. 2. Multi-layer access controls and MFA. 3. Continuous security monitoring. 4. Annual independent security audits. 5. Secure backup and disaster recovery systems.

PART 12: Awareness and Education

Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

School Programs (Age 6-18)

[Hint: Design age-appropriate curriculum topics, activities, and learning outcomes.]

- Introduce cybersecurity education from primary school level.
- Teach safe internet use, password security, cyberbullying awareness, and online privacy.
- Conduct annual cybersecurity competitions and awareness activities.
- Include practical lessons on identifying phishing and online scams.

College & University Programs

[Hint: Describe cybersecurity modules, certification pathways, and lab requirements.]

- Introduce mandatory cybersecurity modules for all students.
- Establish cybersecurity laboratories and research centers.
- Encourage industry certifications such as Security+, CEH, and CISSP.
- Promote internships and cyber defense competitions.

Corporate Training Programs

[Hint: Mandatory training requirements, frequency, and assessment methods for employees.]

- All employees shall undergo cybersecurity awareness training annually.
- Conduct phishing simulation exercises and security drills.
- Train employees on data protection, incident reporting, and password management.
- Assess employee knowledge through periodic evaluations.

Social Media Campaigns

[Hint: Define platforms, messaging themes, content types (videos, infographics), and target demographics.]

- Launch nationwide campaigns on Facebook, Instagram, LinkedIn, X, and YouTube.
- Publish awareness videos, infographics, and cyber safety tips.
- Promote themes such as phishing prevention, online privacy, and ransomware awareness.
- Collaborate with influencers and educational institutions to increase outreach.

Public Awareness Workshops

[Hint: Location, frequency, target audience, topics, and delivery method for community workshops.]

- Conduct free workshops in schools, colleges, community centers, and government offices.
- Educate citizens on cyber fraud, digital payments, social media safety, and identity theft.
- Organize workshops quarterly in urban and rural areas.
- Provide multilingual awareness materials for wider accessibility.

PART 13: International Cooperation

Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

Cybercrime Investigations

[Describe VIT's policy and procedures for international collaboration on this topic. Reference real-world frameworks like the Budapest Convention as examples if needed.]

VIT's shall cooperate with foreign law enforcement agencies to investigate cross-border cybercrimes such as ransomware, phishing, cyber fraud, and cyber terrorism. Requests for digital evidence, suspect identification, and forensic assistance shall be processed through mutual legal assistance treaties (MLATs) and international agreements.

Information Sharing Agreements

[Describe VIT's policy and procedures for international collaboration on this topic. Reference real-world frameworks like the Budapest Convention as examples if needed.]

VIT's shall establish agreements with international cybersecurity agencies, CERT organizations, and trusted partner nations to exchange information on cyber threats, vulnerabilities, malware campaigns, and emerging attack techniques. Shared information shall be protected under confidentiality and data protection laws.

Joint Operations

[Describe VIT's policy and procedures for international collaboration on this topic. Reference real-world frameworks like the Budapest Convention as examples if needed.]

The NCSA may participate in joint cybercrime investigations and multinational cybersecurity exercises with INTERPOL, CERT organizations, and partner countries. Joint operations may target cybercriminal groups, ransomware networks, financial fraud syndicates, and critical infrastructure threats.

Extradition Requests

[Describe VIT's policy and procedures for international collaboration on this topic. Reference real-world frameworks like the Budapest Convention as examples if needed.]

Individuals involved in serious cybercrimes against VIT's may be extradited under applicable treaties and international law. Extradition requests shall follow due legal process and respect human rights obligations while ensuring accountability for cyber offenses.

Threat Intelligence Exchange

[Describe VIT's policy and procedures for international collaboration on this topic. Reference real-world frameworks like the Budapest Convention as examples if needed.]

The NCSA shall exchange real-time threat intelligence with international cybersecurity partners to identify emerging threats, malware campaigns, and nation-state cyber activities. Information sharing shall support early warning systems, coordinated responses, and protection of critical infrastructure.

PART 14: Case Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: National Citizen Database Breach

Crime Category	Unauthorized Access / Data Breach
What Happened	A hacker exploited an unpatched government server and gained unauthorized access to the National Citizen Database of VIT's. Personal information of nearly 500,000 citizens, including names, addresses, and identification numbers, was stolen and later sold on the dark web.
Law Violated	Unauthorized Access, Data Theft, Privacy Protection Act, and Data Security Requirements.
Investigation Process	The NCSA collected server logs, traced the attacker's IP addresses, performed digital forensic analysis, and coordinated with internet service providers. The attacker was identified and arrested.
Punishment Given	7 years imprisonment and a fine of ₹25 lakh.
Lessons Learned	Government agencies must implement timely patch management, regular security audits, and stronger access controls.

Case Study 2: VIT National Bank Phishing Scam

Crime Category	Financial Fraud / Phishing
What Happened	Cybercriminals created a fake banking website identical to the official VIT National Bank portal. Thousands of customers received phishing emails requesting account verification. Over 2,000 citizens entered their credentials, resulting in financial losses exceeding ₹10 crore.
Law Violated	Phishing, Identity Theft, Financial Fraud, and Unauthorized Data Collection.
Investigation Process	The NCSA worked with the bank, domain registrars, and law enforcement agencies to identify the fraudulent infrastructure. Digital evidence led to the arrest of the operators.

Punishment Given	10 years imprisonment and a fine of ₹50 lakh.
Lessons Learned	Banks should implement stronger customer awareness programs, Multi-Factor Authentication, and anti-phishing monitoring systems.

Case Study 3: Ransomware Attack on Central Hospital

Crime Category	Ransomware / Cyber Terrorism
What Happened	A ransomware group infiltrated the Central Hospital network through a phishing email. Patient records and critical medical systems were encrypted, disrupting emergency healthcare services. Attackers demanded cryptocurrency in exchange for restoring access.
Law Violated	Ransomware Distribution, Unauthorized Access, Critical Infrastructure Protection Law, and Cyber Terrorism provisions.
Investigation Process	The NCSA activated the National Incident Response Team, isolated infected systems, collected forensic evidence, and worked with international cybersecurity agencies to identify the attackers.
Punishment Given	Life imprisonment and a fine of ₹1 crore.
Lessons Learned	Critical infrastructure organizations must maintain secure backups, employee awareness training, network segmentation, and continuous cybersecurity monitoring.

PART 15: Future Recommendations

Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

AI Security (2025-2035)

[Hint: How should VIT's regulate increasingly powerful AI systems? Consider autonomous AI decision-making, AI-powered cyberattacks, and AI bias.]

VIT's shall require AI systems to undergo regular security and bias audits. High-risk AI applications must include human oversight, transparency, and accountability mechanisms. The use of AI for cyberattacks, deepfake fraud, or automated malicious activities shall be strictly prohibite.

Quantum Computing

[Hint: Quantum computers can break current encryption. How should VIT's prepare? When should organizations upgrade to quantum-resistant cryptography?]

Government agencies and critical infrastructure operators shall begin transitioning to quantum-resistant encryption standards by 2030. Research and investment in post-quantum cryptography shall be encouraged to protect sensitive national data from future quantum threats.

Privacy Protection

[Hint: Propose stronger privacy protections as data collection becomes more pervasive. Consider biometric data, behavioral profiling, and smart device data.]

Future privacy laws shall provide stronger protection for biometric data, behavioral profiling, location information, and smart device data. Citizens shall have greater control over how their personal information is collected, processed, and shared.

Digital Identity

[Hint: Recommend a national digital identity framework — secure, inclusive, and privacy-preserving. Address the risks of centralized identity systems.]

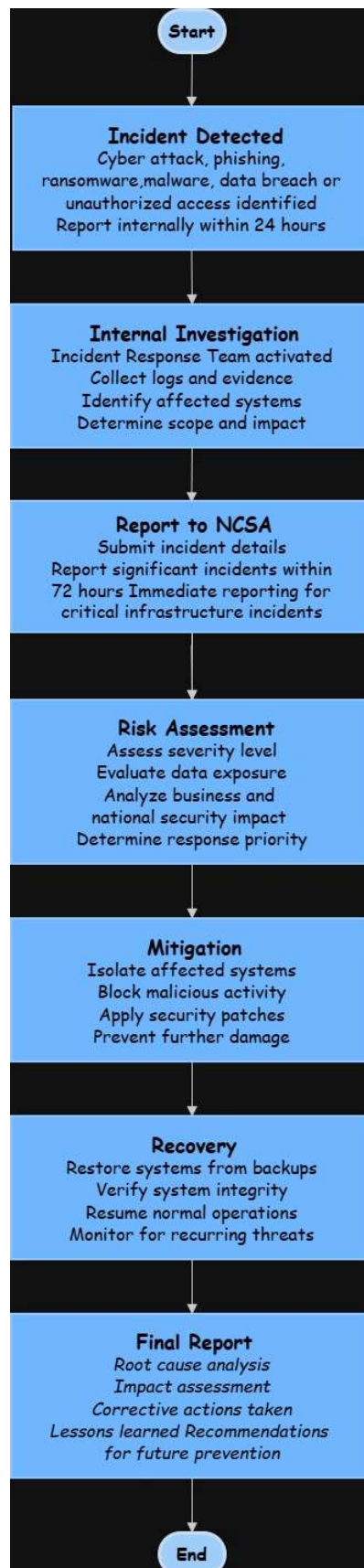
VIT's shall develop a secure national digital identity framework using Multi-Factor Authentication (MFA), encryption, and privacy-preserving technologies. Citizens shall have full visibility into how their identity data is accessed and used.

Cyber Warfare

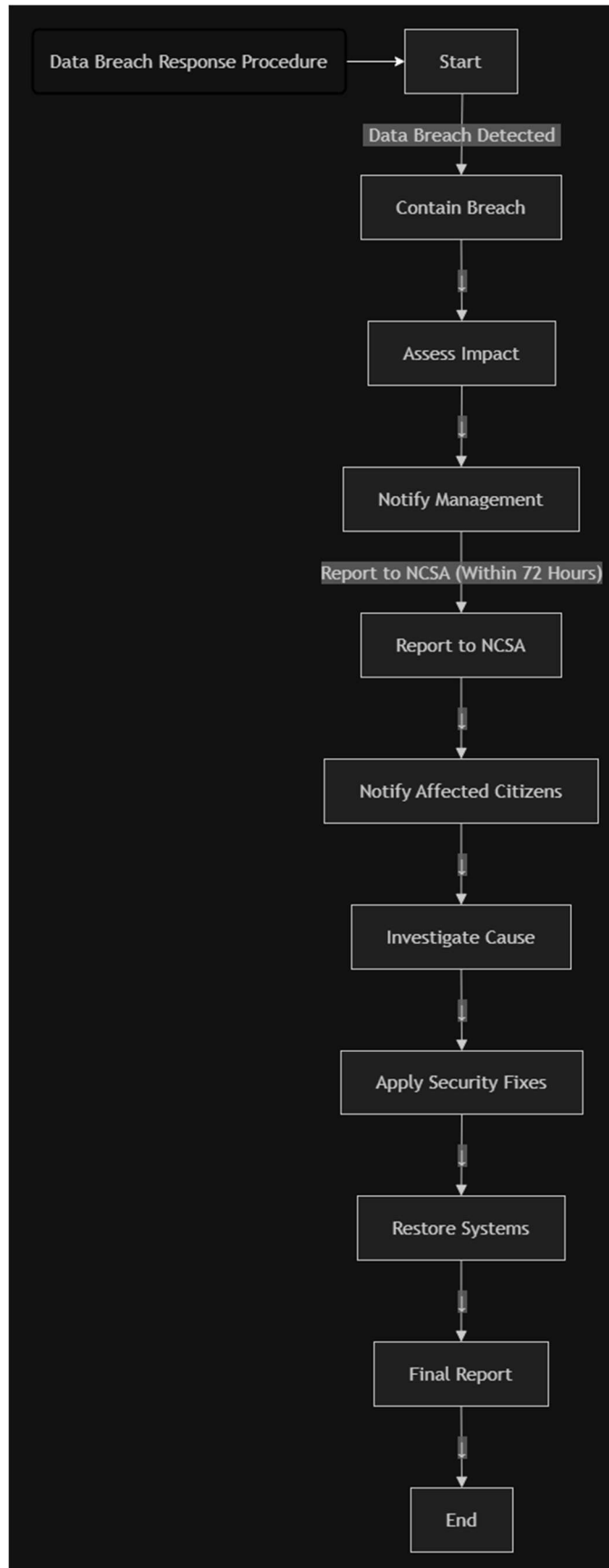
[Hint: How should VIT's law address state-sponsored cyberattacks? Define what constitutes an act of cyber war and VIT's response policy.]

VIT's shall establish a national cyber defense strategy to address state-sponsored cyberattacks and cyber warfare. Critical infrastructure attacks targeting national security, healthcare, energy, or financial systems shall be treated as acts of cyber warfare and responded to through coordinated national and international action.

Cybercrime Reporting Flowchart



Flowchart: Investigation Process



Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Criteria	Max Marks	Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Done	Deliverable	Requirements
☐	Project Report	Minimum 25 pages, well-formatted, all 15 parts completed, diagrams included
☐	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
☐	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution
☐	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment

Done	Deliverable	Requirements
☐	Flowchart: Data Breach Response	Shows data breach response procedure for organizations
☐	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
☐	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
☐	Infographic: Security Framework	Visual overview of the data protection and corporate security framework

Good Luck!

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research real-world cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!